

Synthetic cohort analysis

Example Analytics Team

Five-year mortality analysis

Synthetic data for software testing only. We analysed **12,480** participants, with **847** events over **58,231.5** person-years. The adjusted hazard ratio was **1.43 (95% CI 1.21–1.68)**. A β -blocker dose of 5 $\mu\text{g/mL}$ is included only as a typography sentinel.

Model

$$h(t | x) = h_0(t) \exp(\beta_1 x_1 + \beta_2 x_2)$$

Results

Phenotype	N	Events	HR	95% CI
Cardiometabolic	6240	381	1.00	Reference
Ischaemic/cerebrovascular	3744	312	1.43	1.21–1.68
Respiratory	2496	154	1.17	0.97–1.41

Reproducible analysis

```
fit <- coxph(Surv(time, event) ~ phenotype + age + bmi, data = cohort)
summary(fit)
```

Limitations

These are synthetic estimates.¹ No inference about patients is intended.
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¹Test data created specifically for TypesetLLM evaluation.